

# Offline Legal Translation — Operating Manual

Version 1.2 · 2026-08-01 · EN↔SL (DE later) · hpelitebook8g1i16, Ubuntu 26.04

This document consolidates and replaces the Hardware Assessment, Workflow Architecture, and Addenda 1 and 2. Version 1.2 revises it for the multi-project confidential layout, the encrypted container, and the Claude Code project location. Everything currently believed correct is here.

## 1. What this system is

A local pipeline that produces draft translations of legal documents on one machine, with no network egress, so that a certifying translator edits rather than translates from scratch. It is judged on one number only: whether editing a machine draft takes less time than translating the same page fresh.

Everything is arranged around that measurement. Section 6 is the experiment that decides whether to proceed; nothing beyond it should be built until that number is known.

### 1.1 The four commands you will use most

| Command          | Purpose                                                 |
|------------------|---------------------------------------------------------|
| tr-project       | List, create, or switch the active confidential project |
| tr-inventory     | Classify every file in source/ by source language       |
| tr-status        | What is translated, what is missing, what is stale      |
| tr-run           | Translate everything in source/ not yet in translated/  |
| tr-lint          | Deterministic checks; produces the reviewer worklist    |
| tr-xlsx --survey | Measure spreadsheet redundancy before committing to it  |

Command blocks are colour-coded: **command** *--option* \$VARIABLE "string" # comment

## 2. Layout and naming

Confidential work lives in one encrypted container holding many projects. Each project is self-contained: its own sources, translations, working files, logs, and translation memory. Filenames are preserved exactly from source to output, so finding gaps is a plain set difference between two directories.

```
~/Claude_Stuff/cli_projects/
translation-tools/      the kit. Outside the data tree.
  bin/ lib/ glossary/ prompts/
~/.translate-venv/      python env. Survives unmount.

~/translation-work/
confidential-projects/  <- ENCRYPTED CONTAINER MOUNTS HERE
.active                which project the tools operate on
_shared/
glossary/base.tsv      terminology reusable across matters
```

|                                     |                                      |
|-------------------------------------|--------------------------------------|
| <b>glossary/nontranslatable.txt</b> |                                      |
| <b>prompts/translate.txt</b>        |                                      |
| <b>kranj-2024/</b>                  | a project                            |
| <b>source/</b>                      | originals. Read-only.                |
| <b>translated/</b>                  | deliverables. SAME filenames.        |
| <b>work/ocr/</b>                    | OCR output and text layers           |
| <b>work/tm.sqlite</b>               | memory, cache, and resume state      |
| <b>logs/</b>                        |                                      |
| <b>glossary/project.tsv</b>         | case-specific terms; override base   |
| <b>project.conf</b>                 | language pair, suffix, OCR languages |
| <b>maribor-2025/</b>                | another project, fully isolated      |

**What is shared and what is not.** Glossaries layer: the shared base supplies settled legal terminology, and each project overlays case-specific renderings that win on conflict. Translation memory does NOT layer. Each project keeps its own, because memory holds actual source and target sentences, and those must not cross between matters with different clients and different confidentiality obligations.

Subdirectories under a project's `source/` are mirrored into `translated/`. If a client permits the suffix variant, set it in that project's `project.conf` and every tool follows:

```
TR_SUFFIX=_translated      # in <project>/project.conf
```

## 2.1 The \$KIT shorthand

The kit path is long, so this manual uses `$KIT` throughout. Define it once and the command blocks below can be copied verbatim.

```
echo 'export KIT=~/.Claude_Stuff/cli_projects/translation-tools' >> ~/.bashrc
source ~/.bashrc
```

The kit deliberately sits on a different branch of the filesystem from the case data. The Claude Code guard refuses to start a session from anywhere inside or above `~/translation-work/`, so the scripts must live outside that tree for `cd $KIT && claude` to be a safe operation. Keeping them under `Claude_Stuff/cli_projects/` also puts all Claude Code work in one place.

No script in the kit hardcodes its own location; each derives it at run time, and `core.hooksPath` is relative, so both travel with the directory. Two lines in `~/.bashrc` do not: the `export KIT=` and the `$KIT/bin PATH` entry `tr-setup` wrote with the old absolute path. Re-running `tr-setup` after a move appends the new `PATH` line but leaves the stale one, so edit `~/.bashrc` by hand.

## 2.2 Where these documents live

Keep the reference documents at `~/translation-work/docs/`, one level above the container — not inside `confidential-projects/`.

The reason is that `confidential-projects/` is a mount point. When the container is closed that directory is empty by design, so documents stored there vanish exactly when they are most needed: at the moment you are trying to remember how to open the container. Documents kept one level up remain available whether the container is open or closed, survive recreating

it, and contain no case material, so nothing is lost by holding them outside the encrypted volume.

```
mkdir -p ~/translation-work/docs  
# then save the manual, handover, and container documents there
```

The Claude Code deny rules are scoped to confidential-projects/ rather than all of ~/translation-work/, so a session can still read these documents while remaining unable to read any case data.

## 2.3 Selecting a project

Every tool operates on the active project. This is the one new way the layout can go wrong, so the tools are deliberately noisy about it: tr-run prints a banner and asks for confirmation before touching anything, and any tool refuses outright when no project is set.

```
tr-project           # show active, list all with file counts  
tr-project kranj-2024    # switch  
tr-project --new maribor-2025 # scaffold and switch  
tr-project --none      # clear  
  
# override for a one-off, without changing the active project  
TR_ROOT=~/translation-work/confidential-projects/kranj-2024 tr-status
```

The active project is recorded in a file inside the container, not in your shell, so it persists across terminals and is unset automatically whenever the container is closed.

### 3. One-time setup

Run these once, in order, from your home directory. Total time is dominated by the model download.

#### 3.1 Install the kit

```
# a shorthand worth putting in ~/.bashrc, used throughout this manual
export KIT=~/.Claude_Stuff/cli_projects/translation-tools

mkdir -p "$KIT"
git clone git@github.com:hmbseaotter/offline-translation-kit.git "$KIT"
chmod +x "$KIT"/bin/*
"$KIT"/bin/tr-setup
source ~/.bashrc           # picks up PATH and TR_PROJECTS
```

tr-setup installs OCR and PDF tooling, builds a Python virtual environment outside the container so it survives unmounting, and warns if swap is too small. It creates no project data — that comes from case-init and tr-project. It is safe to re-run.

Then the container and the guard, both covered in full by the Encrypted Case Container document:

```
cd "$KIT"
./bin/case-init 40G           # LUKS2 container, once

cp bin/case-guard ~/.local/sbin/claude # refuses to start while mounted
chmod +x ~/.local/sbin/claude

case-open
tr-project --new kranj-2024
```

#### 3.2 Increase swap

Do this before any long run. It is a safety net against the out-of-memory killer terminating a multi-day job, not a performance measure; inference should never actually touch swap.

```
sudo swapoff /swapfile
sudo fallocation -l 8G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile && sudo swapon /swapfile
free -h           # expect 8.0Gi swap
```

#### 3.3 Install and register the model

GaMS3-12B-Instruct is a Slovene-adapted model from the University of Ljubljana, built on Gemma 3. It is not in the Ollama library, so it comes from Hugging Face as a GGUF. Q8\_0 is chosen because quantization damage falls hardest on exactly the sparse morphological patterns Slovene depends on, and throughput is not a constraint here.

```
tr-model           # ~13 GB download, once

# equivalent to:
```

```
# ollama pull hf.co/mradermacher/GaMS3-12B-Instruct-GGUF:Q8_0
# ollama create gams3:q8 -f <Modelfile pinning num_ctx and temperature>

ollama list           # confirm gams3:q8 is present
```

The alias matters. Every script refers to gams3:q8, so switching models later is one tr-model invocation rather than an edit in eight places. It also pins num\_ctx and temperature so behavior does not drift with an Ollama upgrade.

**Verify the tag at pull time.** The repository publishes both static and imatrix-weighted quantizations and tag syntax occasionally changes. If tr-model fails, open the model page, copy the exact tag, and pass it as the first argument: tr-model hf.co/...:Q8\_0

### 3.4 Confidentiality hardening

The model makes no network calls, but two exposures remain and both are more likely to cause a breach than the inference stack.

Confirm the disk is encrypted. Case material on a laptop is exposed by theft, not by the network:

```
lsblk -o NAME,FSTYPE,MOUNTPOINT | grep -i crypt
```

If that returns nothing, the installation is unencrypted. Retrofitting is disruptive; an encrypted container holding only ~/translation-work/confidential-projects — which is what case-init creates — is the pragmatic alternative.

Confirm no cloud-routed models. Recent Ollama versions can dispatch to hosted models, which would send document text off the machine:

```
ollama list           # no "-cloud" tagged entries
ollama ps             # what is actually loaded right now
```

### 3.5 Verify the install

```
echo "Obdolženi je bil obtožen po čl. 211 odst. 2 KZ-1." \
| ollama run gams3:q8 --verbose
```

Read two things from the --verbose output: prompt eval rate (prefill) and eval rate (generation). Measured on this hardware with gams3:q8, over 12 samples of real pipeline traffic: prefill 7.00 tokens per second, generation 2.20. What governs planning is neither of those but the sustained rate — 0.81 output tokens per second, 48.3 seconds of wall clock per segment. The earlier estimate of 4 tokens per second was optimistic by roughly five times, because most of a segment's cost is not generation: every call re-reads a 200-token system prompt at 7 tokens per second, some 28 seconds, and pays it whether the segment is a sentence or two words. A faster quantization therefore attacks the smaller half of that cost; batching several segments into one call, or shortening the system prompt, attacks the larger.

## 4. Before bulk translation

Four things to settle before the corpus is processed. The first decides which files are in scope at all. The other three determine output quality far more than any model setting, and none of them can be automated.

### 4.1 Triage — which files are in scope

A client drop is not a clean corpus. Files arrive in the folder structure the client used — often two or more separate drops, nested several levels — and that structure is carried through source/ to translated/ untouched. The tree also mixes languages: Slovene alongside English, Croatian or Serbian, Armenian, and whatever else a matter happens to contain. Only the files in the project's source language are translated; the rest are counted and set aside.

tr-inventory walks source/ in full, records every file whether or not the pipeline can convert it, and detects the language of each. Run it before tr-status or tr-run — both work from its manifest.

```
tr-inventory      # classify every file in source/
tr-inventory --rescan  # re-examine files already recorded
tr-inventory --no-ocr # skip the sampling pass on scanned PDFs
```

This is not tidiness. A Croatian file pushed through a Slovene-to-English prompt costs hours of inference at 48 seconds a segment and produces confident nonsense — and the result is written into work/tm.sqlite keyed on the source text, so it is reused silently every time that segment reappears. Undoing it means editing the memory or retranslating the matter. tr-run warns rather than proceeding quietly when no inventory exists.

Three files are written to work/inventory/. manifest.tsv lists every file with its path, format, detected language and confidence. by-lang/ splits those paths into one list per language. summary.txt holds counts and no paths at all, and is the only one of the three that may be sent to the client — a filename in a criminal matter routinely carries a party name, a date or a case number.

Scanned PDFs have no text layer to detect from, so tr-inventory OCRs a page or two with the candidate languages combined and decides from that; the full OCR pass in S5.3 then runs in the right language rather than a guessed one. Anything it cannot place — too little text, an unsupported format, an unreadable file — goes to the undetermined bucket with its path, for a human to deal with. Nothing is guessed.

### 4.2 The glossary — highest leverage

Two files, layered. The shared base holds legal terminology that carries across matters; the project file holds case-specific renderings and wins on conflict. The certifying translator fixes each once, up front, and every later deviation is reported by tr-lint with its location.

```
P=~/.translation-work/confidential-projects

nano $P/_shared/glossary/base.tsv      # reusable across cases
nano $P/kranj-2024/glossary/project.tsv # this matter only
```

```
# obdolženi<TAB>the accused<TAB>criminal procedure
# oškodovanec<TAB>the injured party
# zaslišanje<TAB>examination<TAB>not "hearing"
```

Promote a term to the base file only once it is settled and not case-specific. Names, entities, and matter-specific usages stay in the project file.

Extract candidates from the corpus rather than inventing them. Charges, statutory references, court and agency names, procedural terms, party designations, document types. The already-installed Qwen model is fast and well suited to drafting the candidate list for the translator to rule on.

### 4.3 Non-translatables

Regex patterns in nontranslatable.txt mark strings reproduced verbatim: case numbers, file references, statute short forms. Defaults for Slovene court formats are supplied. Every one of these the model would otherwise “helpfully” translate is a guaranteed edit at a known location, so catching them in code is free quality.

Dates, amounts and times are NOT in that class. They are converted, because that is what a translation is. Into English, 5. 3. 2024 becomes March 5, 2024; 12.450,00 becomes 12,450.00; a 24-hour time becomes 2:30 p.m. Into Slovene it runs the other way: 5. marec 2024 - the day carries a period because it is an ordinal, so bare “5” would read as the cardinal number rather than the fifth day; the month name is lowercase, which Slovene requires; and the three parts are spaced. Amounts take 12.450,00 and times stay on the 24-hour clock. The value never changes; only its spelling does. Two forms are wrong in either language and worth naming, because both were in this kit before the translator corrected them: “5 March 2024” and “5. Marec 2024”. The all-numeric 5.3.2024 is technically correct Slovene but invites the reader to wonder whether it means 5 March or 3 May, which is a poor property for evidence.

The distinction has a practical edge. Inside a sentence the model does the conversion. A segment that is only a date or an amount never reaches the model at all, because the patterns above exclude it from translation - so a Datum column in a spreadsheet stayed in Slovene while the same date in prose came out in English, and the deliverable contradicted itself column by column. Those whole-segment values are now converted in code: exact, consistent, and without the 48 seconds a model call would cost.

```
nano $P/_shared/glossary/nontranslatable.txt

# test a pattern before trusting it (kit dir, no case data needed):
cd "$KIT" && python3 -c "
import sys; sys.path.insert(0,'lib'); import trlib
print(trlib.is_translatable('I K 12345/2024')) # expect False"
```

### 4.4 Seed the memory with the translator’s prior work

The metric is edits this particular translator makes. If any prior EN↔SL legal translations exist, aligning them into the memory means drafts arrive already using their established

renderings and register. A draft that matches their habits produces fewer corrections than an objectively similar draft that does not. This is probably the single highest-value input available and costs one conversation plus an alignment pass.

## 5. Handling each source type

### 5.1 Word documents

```
cd ~/translation-work/confidential-projects/kranj-2024
tr-docx source/ovadba.docx translated/ovadba.docx --from sl --to en
```

Paragraphs, headings, lists, tables, headers and footers are preserved. Each paragraph is split into sentences before translation, so the memory keys on sentences and repeats are reused across the whole corpus.

**Known limitation.** Formatting that varies within a single paragraph — one bold word mid-sentence — is not preserved; the paragraph takes the formatting of its first run. Paragraph-level styling is intact. If intra-paragraph emphasis is legally significant in this corpus, use OmegaT for those files instead (Section 9).

### 5.2 Spreadsheets

Always survey first. Large tables are dominated by repeated categorical values, and deduplication both collapses the work and guarantees identical cells receive identical translations.

```
# measure first, translate nothing
tr-xlsx source/tabela.xlsx --survey

# then translate; each distinct string is translated once
tr-xlsx source/tabela.xlsx translated/tabela.xlsx

# restrict to specific columns if only some hold text
tr-xlsx source/tabela.xlsx translated/tabela.xlsx --cols B,D,E
```

The survey prints the unique-string ratio and an estimate of translation time with and without deduplication. On the 7,000-row table, run this before planning anything around it. Output stays in XLSX, which is correct — ten columns will not transfer usefully into Word.

### 5.3 Scanned PDFs — two stages, deliberately

This is the highest-risk path in the system, for a specific reason: a model fed corrupted OCR does not flag the corruption. It produces a fluent, confident translation of the corruption. A misread digit in a date or an amount arrives looking entirely correct.

```
# stage 1 - OCR only, then stop
tr-pdf --ocr-only source/zapisnik.pdf

# read the text layer against the page images
less work/ocr/zapisnik.txt
```



```
xdg-open work/ocr/zapisnik.ocr.pdf
```

```
# stage 2 - only after verification
```

```
tr-pdf source/zapisnik.pdf translated/zapisnik.docx
```

Check names, dates, case numbers, amounts, and the characters č, š, ž specifically. Mangled diacritics can still produce valid Slovene words, so no spell-check will catch them. OCR output is cached, so stage 2 does not re-run it.

A second engine is worth adding if scan quality is poor. A vision model reads a page by a completely different mechanism than Tesseract, so their failure modes are largely uncorrelated. Run both, compare only the numbers, dates, and capitalized tokens, and inspect where they disagree.

Vision input works. The installed Qwen model transcribed a rendered Slovene legal page at 97.3% against the page's own text layer, with every case number, amount, date and diacritic exact. The 2.7% shortfall was the model being right where the comparison was wrong: it read the evidence table as a table, keeping Datum with its date and Znesek with its amount, where pdftotext flattened the headers into one column and the values into another. For a document where a figure must stay bound to its label, that is the difference that matters.

It costs about 6.7 minutes a page, against seconds for Tesseract. So the cross-check is affordable on the pages that carry the risk - those with case numbers, dates and amounts, or where Tesseract's own confidence is low - and not affordable across a whole corpus. Which pages is the open question; whether it works is not.

One caveat before relying on it: the page tested was a clean render, not a scan. Skew, speckle, stamps and handwriting are untested.

```
ollama run qwen3.6 "Transcribe this page verbatim." ./page.png
```

## 6. Phase 1 — the experiment that decides everything

Do not build further until this number exists. The comparison is deliberately simple.

| Step | Action                           | Detail                                                                                                   |
|------|----------------------------------|----------------------------------------------------------------------------------------------------------|
| 1    | Pick six comparable pages        | Three for machine assistance, three matched for scratch translation. Same document type, similar density |
| 2    | Draft the three                  | tr-run, or tr-txt --bilingual for side-by-side review                                                    |
| 3    | Time the translator on both sets | Minutes per page, MT-assisted versus from scratch                                                        |
| 4    | Compute the ratio                | This is the decision. Anything at or above 1.0 means stop                                                |
| 5    | Ask which errors cost the time   | Terminology? Register? Numbers? That tells you what to fix next                                          |

```
case-open && tr-project kranj-2024
cp /path/to/test/*.docx \
~/translation-work/confidential-projects/kranj-2024/source/
tr-status
tr-run
tr-lint

# bilingual layout for side-by-side review (source in grey above target)
tr-txt work/ocr/sample.txt /tmp/sample-bilingual.docx --bilingual
```

Step 5 matters as much as step 4. A ratio slightly above 1.0 caused by one fixable error class — say, untranslated case numbers — is a different verdict than the same ratio caused by systematically wrong register.

## 7. Daily operation

```
case-open      # unlock the container
tr-project     # confirm which project is active
tr-inventory   # classify the drop; after new files arrive

tr-status      # what remains
tr-status --missing # bare paths, for piping
tr-status -v    # include the completed list

tr-run -n      # dry run: what would be processed
tr-run         # everything missing
tr-run source/one.docx # a single file

tr-lint        # after any run

case-close     # lock up when finished
```

tr-run is resumable at two levels. It skips files already present in translated/, and within a file every segment already in the memory is reused rather than regenerated. Interrupting it costs at most one segment. Re-running after editing the glossary is cheap for the same reason.

A file is re-translated only if its source is newer than its output. To force a redo, delete the output. To force a redo ignoring the memory, change TR\_PROMPT\_VERSION, which is part of the cache key:

```
rm translated/ovadba.docx && tr-run # reuse memory
TR_PROMPT_VERSION=v2 tr-run # ignore memory, retranslate
```

## 8. The lint report

tr-lint runs no model, takes seconds, and catches the error classes a language model is structurally worst at noticing. Its output is not a pass/fail gate; it is a prioritized worklist to hand the translator alongside the drafts.

| Check | Meaning                                                                                                                                       |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| FAIL  | The segment errored out and contains no translation. Fix and re-run first                                                                     |
| NUM   | A number in the source is absent from the target, or a number appears that was not in the source. Highest consequence class in legal evidence |
| NONTR | A string marked non-translatable was altered or dropped                                                                                       |
| INCON | The same source sentence was translated two different ways. A corpus-level finding no per-document review will surface                        |
| GLOSS | An agreed glossary term was not used                                                                                                          |
| ECHO  | Target identical to source — possible untranslated passthrough                                                                                |
| EMPTY | Empty target                                                                                                                                  |

```
tr-lint
tr-lint -tsv work/for-translator.tsv

# open the report in a spreadsheet for the reviewer
libreoffice --calc work/lint-report.tsv
```

Findings are sorted by severity, so the top of the file is where to start. Numeric comparison normalizes separators, so 1.234,56 and 1,234.56 are treated as the same number, and it folds locale conversions together as well: a month that became a word and an hour that shifted by twelve are the same value written differently, not a missing number. Without that, every date in the corpus raised a finding, and legal documents are made of dates.

## 8.1 The error the linter cannot catch

Give the model a source that quotes a provision and stops short of the famous continuation, and it finishes the sentence from memory. Asked to translate “everyone is entitled to a fair hearing before an independent and impartial tribunal”, it returns that plus “established by law” - three words the source does not contain. That is an interpretation presented as a translation, and in a certified document it is the translator's name on it.

Two properties make this the worst class of error in the system. It reads perfectly, so nothing draws the eye to it. And it carries no number, no glossary term and no non-translatable fragment, so no deterministic check can see it - every check in the table above passes.

Nor is it a prompt problem. Four prompt variants were tried, including one whose rule named the failure and explained why it is wrong; the model complied with everything else in that prompt and completed the provision anyway. Measured over ten well-known provisions, two were completed - both ECHR article 6, the most quoted text in criminal procedure.

What can be detected is the context. Any segment citing a statute or treaty is flagged, and those are the segments to read against the source word by word. Asking the model to audit its own output against the source caught both real additions with no false alarms, at about 12 seconds a segment - a second opinion is available if wanted, and applied only to flagged segments it costs a few percent of a run rather than several times it.

The gap that remains: a famous provision paraphrased without a citation marker is not flagged, and nothing detects it.

## 9. When to use OmegaT instead

The scripts above are the fast path: they produce deliverables directly, with filenames preserved, which is what the client wants. They are the right tool for Phase 1 and for the spreadsheet.

OmegaT is the better container for sustained production work, for three reasons: it preserves intra-paragraph formatting properly, it gives the translator a sentence-by-sentence review interface with source and target aligned, and it accumulates a translation memory they own and can reuse on later matters.

The two coexist. Pre-translate with these scripts, export the memory as TMX, and import it into an OmegaT project so segments arrive pre-populated. That path requires no plugin and works with any CAT tool the translator prefers, including Trados.

```
sudo apt install omegat

# export the accumulated memory as TMX for import
cd "$KIT" && python3 -c "
import sys,sqlite3,html; sys.path.insert(0,'lib'); import trlib
db=sqlite3.connect(trlib.path('work','tm.sqlite'))
rows=db.execute('SELECT src,tgt,direction FROM tm').fetchall()
print('<?xml version=\'1.0\' encoding=\'UTF-8\'?>')
print('<tmx version=\'1.4\'><header srclang=\'sl\' '
      'segtype=\'sentence\' datatype=\'plaintext\'><body>')
for s,t,d in rows:
    a,b=d.split('-')
    print(f'<tu><tuv xml:lang=\'{a}\'><seg>{html.escape(s)}</seg></tuv>'
          f'<tuv xml:lang=\'{b}\'><seg>{html.escape(t)}</seg></tuv></tu>')
print('</body></tmx>')" > work/export.tmx
```

*The TMX export above is a sketch and should be validated against OmegaT before relying on it.*

## 10. Long unattended runs

A job running for days has failure modes a short one does not.

```
# prevent suspend from silently halting the run
systemd-inhibit --what=idle:sleep:handle-lid-switch \
               --why="translation batch" tr-run

# or run detached and reattach later
sudo apt install tmux
tmux new -s tr
tr-run          # then Ctrl-b d to detach
tmux attach -t tr

# monitor from another terminal
watch -n5 "sensors | grep -i 'Package id'; free -h | head -2; ollama ps"
tail -f "$(tr-project | awk '/^active/{print $2}')" /logs/run-*.log
```

Keep the machine on mains power and elevated for airflow. A 15 W mobile processor at sustained full load for days is outside its normal duty cycle. If the firmware exposes a charge limit, cap it so the battery is not held at full charge continuously:

```
cat /sys/class/power_supply/BAT0/charge_control_end_threshold
echo 80 | sudo tee /sys/class/power_supply/BAT0/charge_control_end_threshold
```

## 11. Troubleshooting

| Symptom                          | Cause and fix                                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| no active project                | Run <code>tr-project &lt;name&gt;</code> . Tools refuse rather than guess                                                     |
| container not mounted            | Run <code>case-open</code> . When closed the path is empty by design                                                          |
| Wrong project translated         | <code>tr-run</code> prints a banner and asks first; read it. Delete the wrong outputs and re-run                              |
| Connection refused               | Ollama is not running. <code>systemctl status ollama</code> ; <code>sudo systemctl start ollama</code>                        |
| [TRANSLATION FAILED] in output   | Model unreachable or timed out. Fix the cause, then re-run; cached segments are not regenerated                               |
| Process killed mid-run           | Out of memory. Check swap (S3.2) and lower <code>TR_NUM_CTX</code>                                                            |
| Output contains commentary       | Model ignored the output-only instruction. Lower temperature, or tighten prompts/translate.txt                                |
| Sentences split at abbreviations | Add the abbreviation to <code>ABBREV</code> in <code>lib/trlib.py</code> and re-run with a new <code>TR_PROMPT_VERSION</code> |
| Case numbers being translated    | Add a pattern to <code>nontranslatable.txt</code> ; verify with <code>is_translatable()</code> per S4.3                       |
| Very slow                        | Confirm mains power and that the platform profile is not power-saver: <code>cat /sys/firmware/acpi/platform_profile</code>    |
| Diacritics wrong in OCR          | Confirm <code>-l slv+eng</code> was used. Set <code>TR_OCR_LANGS</code> if the pair differs                                   |

## 12. Environment variables

| Variable                                  | Default                                                    | Purpose                                                                    |
|-------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------|
| <code>TR_PROJECTS</code>                  | <code>~/translation-work/<br/>confidential-projects</code> | Container root holding all projects                                        |
| <code>TR_ROOT</code>                      | (active project)                                           | Override to target one project for a single command                        |
| <code>TR_MODEL</code>                     | <code>gams3:q8</code>                                      | Model alias used by every script                                           |
| <code>TR_SRC</code> / <code>TR_TGT</code> | <code>sl</code> / <code>en</code>                          | Per-project, in <code>project.conf</code> . Use <code>de</code> for German |
| <code>TR_SUFFIX</code>                    | (empty)                                                    | Per-project, in <code>project.conf</code> . Set if the client requires it  |
| <code>TR_NUM_CTX</code>                   | <code>8192</code>                                          | Context window. Lower if memory is tight                                   |
| <code>TR_PROMPT_VERSION</code>            | <code>v1</code>                                            | Part of the cache key. Bump to force retranslation                         |
| <code>TR_OCR_LANGS</code>                 | <code>slv+eng</code>                                       | Tesseract languages. Add <code>deu</code> for German                       |

| Variable            | Default                                  | Purpose                                                                              |
|---------------------|------------------------------------------|--------------------------------------------------------------------------------------|
| TR_OLLAMA           | http://127.0.0.1:11434                   | Ollama endpoint                                                                      |
| TR_VENV             | ~/translate-venv                         | Python environment the scripts re-exec into. Set before tr-setup to put it elsewhere |
| TR_NO_REEXEC        | (unset)                                  | Set to 1 to stay on the system interpreter. Diagnostics only; imports will fail      |
| CASE_IMG            | ~/case/confidential.luks                 | The LUKS container file. Read by case-init, case-open, case-status                   |
| CASE_MAP            | casedata                                 | Device-mapper name while the container is unlocked                                   |
| CASE_MNT            | ~/translation-work/confidential-projects | Where the container mounts. Also what the claude guard checks                        |
| TR_DICTS            | /usr/share/hunspell                      | Where tr-inventory looks for the hunspell word lists it detects language with        |
| TR_OCR_SAMPLE_LANGS | slv+hrv+eng                              | Tesseract languages for the detection sampling pass on scanned PDFs                  |

For a German matter, set the pair once in that project's project.conf rather than on the command line, so every later run inherits it:

```
tr-project --new berlin-2026
nano ~/translation-work/confidential-projects/berlin-2026/project.conf

TR_SRC=de
TR_TGT=en
TR_OCR_LANGS=deu+eng
```

German is high-resource and well served by base Gemma; GaMS3 continual pre-training on Slovene may have eroded it. Compare against gemma3:12b on a German sample before relying on GaMS3 for that pair.

## 13. Decisions and why

| Decision              | Chosen             | Reason                                                                                                   |
|-----------------------|--------------------|----------------------------------------------------------------------------------------------------------|
| Model                 | GaMS3-12B-Instruct | Slovene-specific; outperforms base Gemma 3 12B on EN→SL and rivals GPT-4o on the Slovene arena           |
| Quantization          | Q8_0               | Quantization damage is disproportionate for less-resourced languages; throughput is not a constraint     |
| No RAM upgrade        | 32 GB retained     | Workload is bandwidth-bound. More capacity adds no bandwidth and only enables slower, larger models      |
| Segment granularity   | Sentence           | Matches the translator's review unit and makes the memory reusable across documents                      |
| Abbreviation handling | Custom list        | Default splitters shatter Slovene legal text at št., čl., odst., which fights the reviewer on every page |

| Decision              | Chosen                     | Reason                                                                                                                                                                                                                                |
|-----------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spreadsheet strategy  | Unique-string map          | Collapses the work and guarantees identical cells translate identically                                                                                                                                                               |
| Verification          | Deterministic linter       | Catches numeric and consistency errors that models cannot self-detect; costs nothing to run                                                                                                                                           |
| Back-translation      | Dropped                    | Its value assumed a reviewer reading target-only. Sentence-by-sentence comparison against source catches the same errors directly                                                                                                     |
| Project isolation     | Separate memory per matter | Memory holds real sentences. Sharing it across clients would move content between matters                                                                                                                                             |
| Glossary layering     | Shared base + overlay      | Terminology is reusable; case specifics are not. Layering gets the benefit without the leak                                                                                                                                           |
| Data location         | Encrypted container        | Makes the assistant boundary structural rather than remembered                                                                                                                                                                        |
| Dates, amounts, times | Converted, not verbatim    | The translator's rule. English takes March 5, 2024 and a decimal point; Slovene takes 5. marec 2024 - ordinal period, lowercase month, spaced - and a decimal comma. Whole-segment values are converted in code, without a model call |
| Institution names     | Translated                 | A court's name is not an identifier. Listing them beside case numbers made two models read the rule two ways, one leaving Slovene in the output                                                                                       |
| Completed provisions  | Flag the context           | The model finishes famous provisions from memory. No prompt stopped it and no deterministic check sees it, so segments citing a statute are flagged for word-by-word review                                                           |
| Disk encryption       | Container only, for now    | The root filesystem is plain ext4. The container protects the case material; nothing outside it is protected, which is a live risk rather than a settled one                                                                          |

## 14. Open items

- Measure the 7,000-row spreadsheet with `tr-xlsx --survey`. The unique ratio determines whether it is hours or days.
- Trial the container on a throwaway 1 GB image and exercise open, close, and the guard before real data goes near it. Nothing has been encrypted yet: case-init has never run, so confidential-projects/ is an ordinary directory. Every tool refuses to write there, correctly, until it exists.
- Decide what to do about the unencrypted disk. The finding is settled - the root filesystem is plain ext4 (S3.4) - and the decision is not. The container covers the case material; it does not cover a file copied out for review, a temporary file, or the swap.
- Decide whether the vision model earns a place in the OCR stage. It works - 97.3% on a rendered page, every case number, amount, date and diacritic exact, and it read a table



correctly where the PDF text layer did not (S5.3). It costs about 6.7 minutes a page, so the question is which pages, not whether.

- Test the vision model on a real scan. The page it read was a clean render; evidence arrives skewed, speckled and stamped, and nothing here shows how it behaves on that.
- Establish how often the model completes a statutory provision from memory (S8.1). Two of ten well-known provisions, both ECHR article 6, is enough to know the risk is real and not enough to know its shape.
- Seed the memory from the translator's prior work if any exists (S4.4).
- Verify German quality against base Gemma before extending to that pair (S12).
- Settle the professional questions with the certifying translator. This item gates everything else.